# LLJ Cards: Evaluating Text Summarization

| Subsection | Q# | Answer |
|---|---|---|
| **CONTEXT** | | |
| Application | 1 | Natural Language Generation (NLG) evaluation, specifically text summarization. |
| Application | 2 | Not discussed in original work. |
| Task | 3 | Evaluating the quality of automatically generated summaries of news articles. |
| Task | 4 | Not discussed in original work. |
| Task | 5 | Not discussed in original work. |
| Cases | 6 | News articles from standard summarization benchmarks. |
| Cases | 6.1 | Standard NLG benchmarks. |
| Cases | 6.2 | News summarization. |
| Cases | 6.3 | English language news articles. |
| Cases | 6.4 | Not discussed in original work. |
| Cases | 6.5 | Not discussed in original work. |
| Cases | 6.6 | Not discussed in original work. |
| Stakeholders | 7 | Academic research; authors from Microsoft Azure AI. |
| Stakeholders | 8 | Academic community and NLP practitioners. |
| Stakeholders | 9 | Not discussed in original work. |

| Subsection | Q# | Answer |
|---|---|---|
| **EVALUATION CRITERIA** | | |
| Systematized Concept | 10 | Quality of automatically generated summaries. |
| Systematized Concept | 11 | High-quality summaries that are fluent, coherent, consistent with the source, and relevant. |
| Systematized Concept | 12 | Yes. |
| Operationalization | 13 | Coherence (1-5 scale): the collective quality of all sentences. The summary should be well-structured and well-organized, building coherently from sentence to sentence. |
| Operationalization | 14 | Not justified in original work. |
| Operationalization | 15 | Validated via correlation with human ratings on benchmark datasets. |
| Operationalization | 16 | Yes. Coherence definition is specifically designed for summarization. |
| Operationalization | 16.1 | Yes, through correlation with human judgments on summarization tasks. |
| Limitations | 17 | Not discussed in original work. |
| Limitations | 18 | Not discussed in original work. |

| Subsection | Q# | Answer |
|---|---|---|
| **LLJ PIPELINE** | | |
| Set-Up | 19 | Evaluation. |
| Set-Up | 20 | LLJ (single model judge). |
| Set-Up | 21 | Automated summaries of English news articles. |
| Set-Up | 22 | Pointwise (score 1-5). |
| Model | 23 | |
| Model | 23.1 | ChatGPT. |
| Model | 23.2 | GPT-3.5 (text-davinci-003) and GPT-4. |
| Model | 23.3 | Not reported in original work. |
| Model | 23.4 | Not reported (paper published 2023). |
| Model | 23.5 | GPT-3.5: temperature=0; GPT-4: n=20, temperature=1, top_p=1. |
| Model | 23.6 | Not discussed in original work. |
| Reproducibility | 24 | Yes. |
| Reproducibility | 24.1 | No. |
| Reproducibility | 24.2 | Not reported. |
| Reproducibility | 24.3 | |
| Budget | 25 | |
| Budget | 25.1 | Not reported. |
| Budget | 25.2 | Not reported. |
| Prompt | 26 | |
| Prompt | 26.1 | None specified. |
| Prompt | 26.2 | You will be given one summary written for a news article. Your task is to rate the summary on one metric... |
| Prompt | 26.3 | Coherence (1-5) with DUC-based definition. |
| Prompt | 26.4 | DUC-based coherence definition. |
| Prompt | 26.5 | Chain-of-Thought evaluation steps provided. |
| Prompt | 26.6 | Examples provided in appendix. |
| Prompt | 26.7 | |
| Prompt | 26.8 | Evaluation Form (scores ONLY): - Coherence: [score] |
| Score | 27 | Weighted summation using token probabilities: $score = \sum_{i=1}^{n} p(s_i) \times s_i$. |
| Score | 28 | Direct scoring led to mode collapse at 3 and integer-only outputs; probabilistic scoring yields continuous, fine-grained scores. |

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| **LLJ EVALUATION** | | |
| Evaluation | 29 | Correlation with human ratings on benchmark dataset. |
| Evaluation | 29.1 | Spearman and Kendall-Tau correlation. |
| Evaluation | 29.2 | BERTScore, MoverScore, FactCC, QAGS, USR, UniEval, GPTScore. |
| Evaluation | 30 | Not reported in original work. |
| Evaluation | 31 | Convergent validity via correlation with human judgments. |
| Evaluation | 32 | Yes. G-Eval exhibits bias toward LLM-generated text; closed-source LLMs limit reproducibility; frequent model updates cause inconsistency. |
| Evaluation | 33 | No. |
| Evaluation | 33.1 | |
| Ethical Considerations | 34 | Not discussed in original work. |
| Ethical Considerations | 35 | Not discussed in original work. |
| Ethical Considerations | 36 | Not discussed in original work. |

| Subsection | Q# | Answer |
|---|---|---|
| **LLJ PROTOTYPING** | | |
| Study Set-Up | 37.1 | |
| Study Set-Up | 37.2 | Validating LLJ-based evaluation against human judgments. |
| Study Set-Up | 37.3 | Not reported. |
| Study Set-Up | 37.4 | Microsoft Azure AI researchers. |
| Data | 38 | Used existing benchmark datasets with human ratings. |
| Data | 39 | Not discussed in original work. |
| Design | 40 | GPT-3.5 (text-davinci-003) and GPT-4 with various prompting strategies (no Chain-of-Thought, simple prompting, Chain-of-Thought with detailed instructions). |
| Design | 41 | Yes. Reported correlation with human ratings for each variant. |
| Validation | 42 | Yes. Convergent validity assessed via correlation with human ratings. |
| Validation | 43 | Not discussed in original work. |
| Refinement | 44 | No changes reported. |
| Refinement | 45 | Highest correlation with human ratings. |